Curling - Not For the Faint of Heart

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Abstract

Include your abstract here. Abstracts should typically be between 100-200 words. One option for structuring an abstract is to begin with a sentence on background, then proceed to the objective of the report, your unique contributions to the problem (aka what you did), and briefly summarize results / findings. In this paper we present two partial pooling Hamiltonian Monte Carlo Markov Chain samplers designed to determine the relationship between Power-plays, ends, and play "success", with the odds of winning a curling match. Our goal was to prove some sort of multiplicative relationship between any of these covariates and said odds, but we still are not quite able to infer much from our results, with many of our credible intervals including one within their range.

Background and Objective

Throughout time immemorial, there have been activities that go against the grain of what are considered to be more traditional sports. Whether log rolling, where competitors attempt to not fall off of a spinning log, or even the curling of today, complete with brooms, ice, and stones, as long as you have someone to root for, anything can be a sport! Curling itself has it's own peculiar contexts making up the sport, thankfully not requiring quite so much balance! Whether it comes down to a team using a power-play to move some stones out of the way, to have a better chance at making the shot they need to win, or even the unique system put in place to evaluate how well a curler's shot matches their intended plan, there are many layers and wrinkles to the sport.

We have been charged by our professor to find some relationship between a couple of the more strategy related aspects of the sport. More specifically, we want to take a look at how exactly the odds of winning a match change depending on whether or not a team uses a power-play on a specific end, (Or play, in more general terms) and on top of that, whether or not these odds change based on whether the play made "Failed in it's intended result."

Methodology

This analysis was completed using two Bayesian logistic regression models. Our outcome value across both was a binary indicator of whether the United States' mixed doubles curling team won their match. The likelihood level in both cases was taken to be a Bernoulli distribution that depends of the probability of success. This probability of success was estimated by a logistic regression model whose composition depended on the specifics of the question. Each of the β regression coefficients in the models were assumed to have Normal distributions with hyper-priors. The distribution of the prior means was taken to be Normal, while the precisions were taken to be Gamma distributed. These choices align with the standard approach to Bayesian logistic regression.

Our first research question explores the impact of the use of a power play on the odds success in a match. A formal model statement for this problem is as follows:

Let y_{ij} be a binary indicator of Team USA's success in mixed doubles curling in match $i = 1, ..., n_j$ in competition $j = 1, 2, 3$. We model the probability of victory π_{ij} as a linear function of the presence of a power play within a match, x_{ij},

$$\begin{aligned} y_{ij} \mid \beta_{0j}, \beta_{1j}, x_{ij} &\sim Bernoulli\left(\pi_{ij}\right) \\ \log\left(\frac{\pi_{ij}}{1-\pi_{ij}}\right) &= \beta_{0j} + \beta_{1j} x_{ij} \\ \beta_{0j} &\sim N\left(m_0, s_0^2\right) \\ \beta_{1j} &\sim N\left(m_1, s_1^2\right) \\ m_0 &\sim N\left(-0.3, 1\right) \\ m_1 &\sim N\left(0, 1\right) \\ 1/s_0 &\sim Gamma(1,1) \\ 1/s_1 &\sim Gamma(1,1) \end{aligned} \tag{1}$$

Our hyper-priors were reached by looking at data from the Beijing Olympics as information regarding the intricacies of curling was difficult to acquire from other sources. In this investigation we were unable to observe a difference between the power play and non power play groups. Thus, we operated under the prior assumption that β_1, which represents the difference in logged odds between the groups is 0. The observed win percentage of Team USA in Beijing was 33%. Although all of these matches included Team USA using a power play, we are assuming there's no difference between the groups and as such can use this observed probability to estimate that the log odds of victory for non power play matches is equal to -0.3.

Our second research question also considers the success of both the power and non-power plays. A second formal model statement is as follows:

Once again, let y_{ij} be a binary indicator of Team USA's success in mixed doubles curling in match $i = 1, ..., n_j$ in competition $j = 1, 2, 3$. We model the probability of victory π_{ij} as a linear function of the presence of a power play within a match, x_{ij}, the success of said power play z_{ij}, and the number of successful non power plays in a match, w_{ij},

$$\begin{aligned} y_{ij} \mid \beta_{0j}, \beta_{1j}, x_{ij} &\sim Bernoulli\left(\pi_{ij}\right) \\ \log\left(\frac{\pi_{ij}}{1-\pi_{ij}}\right) &= \beta_{0j} + \beta_{1j} x_{ij} + \beta_{2j} z_{ij} + \beta_{3j} w_{ij} \\ \beta_{0j} &\sim N\left(m_0, s_0^2\right) \\ \beta_{1j} &\sim N\left(m_1, s_1^2\right) \\ \beta_{2j} &\sim N\left(m_2, s_2^2\right) \\ \beta_{3j} &\sim N\left(m_3, s_3^2\right) \\ m_0 &\sim N\left(0, 1\right) \\ m_1 &\sim N\left(0, 1\right) \\ m_2 &\sim N\left(0, 1\right) \\ m_3 &\sim N\left(0, 1\right) \\ 1/s_0 &\sim Gamma(1,1) \\ 1/s_1 &\sim Gamma(1,1) \\ 1/s_2 &\sim Gamma(1,1) \\ 1/s_3 &\sim Gamma(1,1) \end{aligned} \tag{2}$$

In order to progress with this analysis, some changes to the data were required. Initially, the data included one observation for each end (of which there are many in a match). The data was pivoted to

have one observation per match with the information regarding ends within a match moved to individual columns. Further, due to the niche nature of the topic, information regarding our research questions was not readily available. Thus, information regarding the Beijing Olympics was filtered out of our dataset for use as priors. This was thought to be worth it for the purposes of a more informed prior but comes with the drawback of a smaller sample size for analysis.

This analysis was completed in RStudio (version 2026.01.0+392). Packages used include tidyverse (version 2.0.0) and rstan (version 2.32.7).

Analysis and Results

The Impact of a Power Play

The power play is a tactic used in mixed doubles curling where the pre-placed stones are moved out to one of the sides. A team is able to use a power play no more than once per match; its use is, however, not required. We aim to investigate how much of an impact on winning the power play has.

First, we must ensure that our model is appropriate for use. The bulk effective sample size (ESS) of our model is well above the cutoff for all values of β_0 and β_1. Five out of the six ESS numbers are above 900, with the sixth being 830 which is still well above the threshold. With the assurance that our ESSs are sufficiently high, we can look at the $\widehat{R}$ which also indicate that the model is acceptable. The $\widehat{R}$ values range from 1.003219 to 1.004226; all values within that range are acceptable for use.

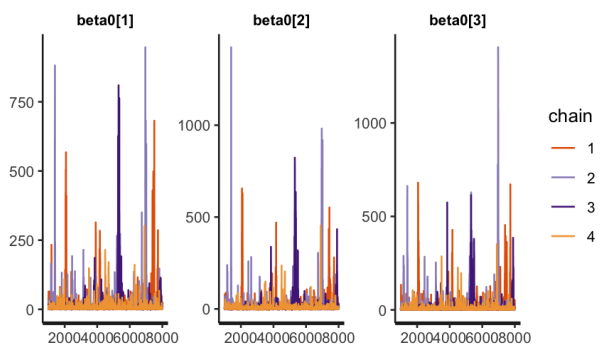

Figure 1: β_0 Traceplots

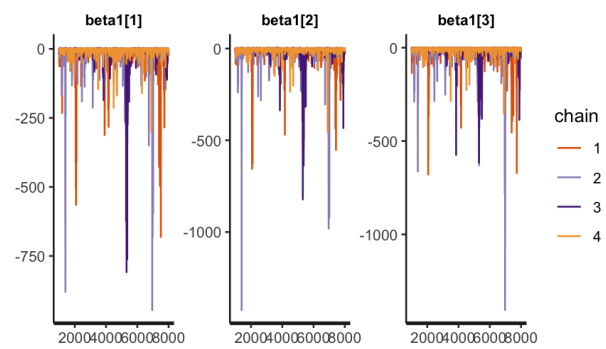

Figure 2: β_1 Traceplots

The traceplots shown above show some problems. The spikes show more deviation from the convergence than we would like. However, when we consider the other convergence metrics they're less concerning than they might me. We do not need to move away from the model, but rather this points to something to consider in future conclusions.

The value of interest in this situation is β_{1j}, which represents the difference in the log odds between matches with power plays and those without for each competition. The values for these βs are -11.48 for $\beta_{1,1}$, -12.25 for $\beta_{1,2}$, and -13.36 for $\beta_{1,3}$. The 95% credible intervals for these βs are: $\beta_{1,1} = [-76.37, 1.05]$, $\beta_{1,2} = [-90.73, 1.46]$, and $\beta_{1,3} = [-93.41, -0.29]$. In order to consider the odds scale, we can exponentiate β_{1j}. Thus, we have $e^{\beta_{1,1}} = 0.42$, $e^{\beta_{1,2}} = 0.79$, and $e^{\beta_{1,2}} = 0.11$. The 95% credible intervals for these values are: $e^{\beta_{1,1}} = [0.00, 2.86]$, $e^{\beta_{1,2}} = [0.00, 4.29]$, and $e^{\beta_{1,3}} = [0.00, 0.75]$.

"How do our intentions affect our winning?"

Now that we have taken a look at how our odds of winning change based on whether a team uses a power-play on a given end, we can take a look at how the "success" of the play dictates a teams odds of

winning. Our trace-plot for our variable of interest from the second Bayesian logistic regression models follows as such:

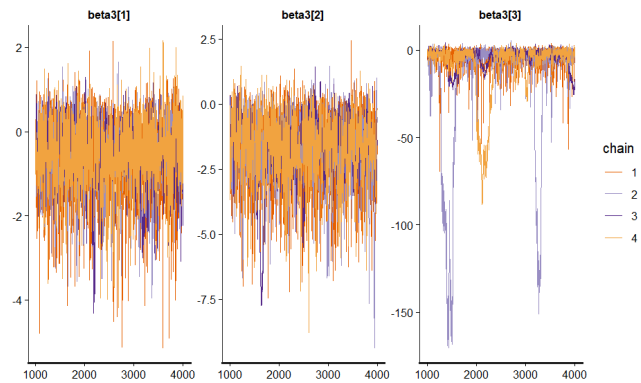

Figure 3: Traceplots for $\beta_{3,1}, \beta_{3,2}$, and$\beta_{3,3}$

While our first two trace-plots appear to be fine, and not have any issues with mixing, our third trace-plot has one or two chains that are deviating just a bit too much from the rest. Moving on to look at our bulk-ESS and $\hat{R}$ might let us look further into this.

When it comes to both $\hat{R}$ and bulk-ESS, our values for the first two years of the competition in our analysis remain reasonable, staying above 400, and below 1.01 respectively. However, when it comes to our $\beta_{3,3}$ coefficient, we have an ESS of 120, and an $\hat{R}$ of 1.07. Perhaps this may be remedied with more samples.

Our value of interest when it comes to this situation is β_{3j}, a variable we coerced to contain the information for the number of successful non-power-plays in a match. Given our regression model, β_{3j} should, after exponentiation, represent the change in log odds of the United States Curling team winning a match, given a one-unit increase in the number of successful non-power-plays in a match, while accounting for the presence of a power-play in a match, as well as the success of said power-play. We derive the following values for $\beta_{3,1}$, $\beta_{3,2}$, and $\beta_{3,3}$:

$\beta_{3,1}$ = -0.64, $\beta_{3,2}$ = -1.81, and $\beta_{3,3}$ = -9.59

With credible intervals of: $\beta_{3,1}$ = [-2.64, 0.72], $\beta_{3,2}$ = [-5.07, 0.18], and $\beta_{3,3}$ = [-88.75, 1.27]

Upon applying an inverse logit function to all coefficient values, to return to an odds scale, we obtain the following coefficients:

$\beta_{3,1}$ = 1.05, $\beta_{3,2}$ = 0.33, and $\beta_{3,3}$ = 0.00013

With credible intervals of: $\beta_{3,1}$ = [0.142, 4.11], $\beta_{3,2}$ = [0.0125, 2.39], and $\beta_{3,3}$ = [5.719983e-39, 7.12]

Discussion

The Impact of a Power Play

Based on the posterior means and 95% we reach different conclusions for different competitions. We cannot make any true conclusions about the World Championships in 2023 and 2024. The credible intervals

for their exponentiated β_1s both include 1, which would indicate no change in odds. This means that the odds of winning seem to be the same in matches where Team USA uses a power play as when they do not. The trend changes when we look at the World Championships in 2025. Here we see that the credible ranges from 0.00 to 0.75. This means that there's a 95% chance that the true multiplicative change in the odds of victory is within those bounds. This would indicate that the use of a power play actually decreases the odds of victory for team USA. Taking all the competitions together, we don't have enough evidence to make a concrete conclusion about the impact of a power play on the odds of winning.

The Impact of Failure

When it comes to how the odds of winning a curling match change based on whether a play "failed in it's intended result," we can look at our coefficients from our second partial pooling model. As all of our coefficients once transformed back to the odds scale contain one within their credible intervals, we have a significant possibility that there is no true change in odds of winning with a given increase in the amount of successful non-power-play plays. This is not to say that the "success" of a play is an irrelevant metric, but perhaps moreso that our model leaves something to be desired. Along with this, the general inconsistency of our $\beta_{3,3}$ coefficient from the 2025 World Curling Championship competition also complicates our analysis, as it does not do well in convergence diagnostics.